

A Unified Starting-Lineup Model for *Marvel Rivals*

Confirmed results

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This report estimates one penalized match-level model of the probability that side 0 wins, conditional on the recorded starting lineups and pre-match information, and derives every reported quantity from that single fit. It replaces separate treatment of designated team-ups, rank bands, and pairwise interactions with one specification in which hero effects, composition effects, hero-by-map deviations, allied pairs, and opposing pairs are estimated jointly, each with a pooled rank slope, under five ridge penalties tuned on chronological development folds. The estimand is an adjusted starting-lineup association: what happens after a team starts a hero, including whatever swaps and responses follow, in the sampled competitive environment. It is not the effect of a hero's abilities in uninterrupted play, and starting attribution alone does not make it a causal estimate.

Sample, regime, and protocol

The development snapshot holds 477,483 competitive PC ranked matches played between 2026-08-07 11:12 and 2026-09-08 11:00 UTC, after the published filters (twelve players, six per side, no draws, scores present, every player with positive play time, at least 240 seconds). The starting record must also be legal: 6,475 matches in which two teammates share the same first-appearance hero are excluded, because a shared start cannot be simultaneous and the record is unverifiable. That leaves 471,008 matches. The official patch record places the whole snapshot inside one balance regime, from the Season 9.5 patch of 7 August to the Season 10 patch of 11 September; the five intermediate updates were cosmetic. A weekly win-rate check flags only The Hood, released on 7 August, whose drift is a learning curve rather than a patch signature.

Collection was stopped at the user's instruction, so the locked confirmation sample is internal: the 19,804 matches played in the final four days of the snapshot, after 2026-09-04 11:00 UTC. The remaining 451,204 development matches feed every design build, fold, tuning run, and table below. The confirmation outcomes are read once, after the candidate is locked, against a predeclared criterion. Because the published headline model was estimated on the whole snapshot, the baseline for that comparison is the identical headline specification refitted on the development rows.

Lobby rank, the mean pre-match score of the twelve starters, runs from 4,141 at the 5th percentile to 4,944 at the 95th, with a median of 4,548. Rank-dependent quantities are evaluated inside that range and no claim is made above 5,200. 14,722 matches have a lobby mean above 5,000, and no individual player in the sample scores above 5,750, so the top of the ranked ladder is outside the data. That is direct coverage evidence.

Specification

Let A_{mh} and B_{mh} indicate whether hero h starts on side 0 or side 1 of match m , r_m the standardized lobby rank, ΔR_m the standardized difference in team mean pre-match score, v_m the map, and $c(\cdot)$ the composition (Tank–Damage–Support counts). With $x_{mh} = A_{mh} - B_{mh}$, $z_{mhk} = A_{mh}A_{mk} - B_{mh}B_{mk}$

and $w_{mhh} = A_{mh}B_{mk} - A_{mk}B_{mh}$ for $h < k$,

$$\begin{aligned} \text{logit } p_m = & \alpha_{v_m} + (\theta_0 + \theta_1 r_m) \Delta R_m + (\gamma_0 + \gamma_1 r_m)^\top [c(A_m) - c(B_m)] + \sum_h (\beta_h + d_h r_m + u_{h,v_m}) x_{mh} \\ & + \sum_{h < k} (S_{hk} + T_{hk} r_m) z_{mhk} + \sum_{h < k} (C_{hk} + D_{hk} r_m) w_{mhk}. \end{aligned}$$

Every block uses the same starting record. The allied block is symmetric and the opposing block antisymmetric, so swapping the two camps negates every feature except the map intercept, which the implementation checks numerically. Every observed composition shape receives its own coefficient, every observed allied and opposing pair its own coefficient, and every hero its own deviation on every map; nothing is removed by a frequency threshold. The blocks are then aliased with each other in known ways: summing a hero's allied features over its five teammates gives five times its own contrast, summing its opposing features over six opponents gives six times it, summing allied features within a role pair gives a function of the composition contrast, and summing hero-by-map deviations over maps gives the hero contrast again. Each candidate centering direction was tested against the data and imposed only when its image lies exactly in the span of the lower-order blocks; 187 of 187 candidates did, with the largest residual at $7.4\text{e-}15$. The reduced design has 6,700 free coefficients.

Five ridge groups shrink the coefficients on their native scale: hero and composition averages (λ_1), allied averages (λ_2), opposing averages (λ_3), hero rank slopes, composition rank slopes and hero-by-map deviations (λ_4), and pair rank slopes (λ_5). Map intercepts and the two rank-imbalance terms are unpenalized. The rank basis and ΔR are scaled with training rows only. Designated team-ups are labels attached after estimation; they receive no separate regressors, constraints, or penalties.

Tuning

Penalties were tuned on three chronological development folds, training on the first 60, 73 and 87 per cent of development matches by play time and validating on the following slice of about 60,000 matches each. A bounded coordinate search on $\log_{10} \lambda$ with warm starts ran 39 evaluations. The selected penalties are $\lambda = (10, 900, 1000, 1000, 30000)$, with mean validation log loss 0.66809 and mean calibration slope 1.002, against 0.67012 for the refitted headline specification on the same folds, a gain of +0.00204. Calibration by supported rank third is 0.984, 1.022, and 0.997.

Table 1: Best evaluations of the penalty search (lowest validation log loss first)

$\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$	Mean validation log loss	Mean calibration slope	Converged
10,900,1000,1000,30000	0.66809	1.002	yes
3.33333,900,1000,1000,30000	0.66809	1.000	yes
30,900,1000,1000,30000	0.66809	1.004	yes
10,900,1000,1000,90000	0.66809	1.002	yes
10,900,1000,1000,1e+09	0.66809	1.002	yes
10,900,1000,1000,10000	0.66811	0.999	yes
10,900,1000,3000,30000	0.66814	1.010	yes
10,900,1000,3000,90000	0.66814	1.010	yes

Table 2: Development ladder: genuine reduced fits at the selected penalties

Model	Free parameters	Fold 1	Fold 2	Fold 3	Mean	Cal. slope
Headline spec., refitted on development rows	187	0.67059	0.67082	0.66896	0.67012	—
Heroes, composition, maps, rank imbalance	96	0.67230	0.67197	0.67030	0.67152	1.009
+ rank slopes and hero-by-map	984	0.67159	0.67118	0.66964	0.67080	1.001
+ allied and opposing pairs (no pair slopes)	3,842	0.66924	0.66814	0.66690	0.66809	1.002
+ pair rank slopes (selected fit)	6,700	0.66923	0.66814	0.66689	0.66809	1.002

Notes. Validation log loss on the same three chronological folds. Reduced fits set the omitted blocks’ penalties to 10^9 , which drives those coefficients to zero, so each row is a genuine reduced model rather than a post-hoc zeroing. The headline row is the published specification (187 parameters, starting-hero attribution, within-role constraint, map intercepts, designated team-ups) refitted on the development rows.

Hero values

The tables that follow come from the selected development fit, estimated on the 451,204 development matches only. Table 3 reports three things per hero. The coefficient β_h is the within-role log-odds contrast at the median lobby rank, reported with its slope per standard deviation of rank, so a reader can place it anywhere in the supported range. The replacement scores are exact predicted probability differences, in percentage points, from placing the hero into a starting slot in place of another hero of the same role, recomputing all five allied and six opposing relationships, the hero-by-map deviation, and the rank slope, and keeping the slot’s player, rank, map, side, and the other eleven starters fixed. Two references are declared. The observed-meta score replaces the hero that actually started in the slot, averaged over 120,000 development matches. The common-reference score compares the hero with the average legal same-role hero in the same slot, so every hero of a role is scored on the same context pool. Legality respects bans and hero uniqueness; coverage is the share of same-role slots in which the hero was a legal replacement, with a median of 88%.

By the common reference, the strongest Tanks are The Thing (+5.0), Peni Parker (+2.3), and The Hood (+2.1), the strongest Damage heroes Magik (+4.4), Black Cat (+4.2), and Storm (+3.0), and the strongest Supports Mantis (+6.2), Rocket Raccoon (+3.9), and Ultron (+3.5). The observed-meta scores by rank third show which heroes’ values move with the environment: Elsa Bloodstone (-1.9 to +0.2), Wolverine (-1.2 to +0.6), and Mantis (+5.8 to +7.2) gain the most from the bottom to the top third, while Black Panther (+0.0 to -2.3), and Cloak & Dagger (+1.5 to -0.9) lose the most. These are pooled slopes evaluated at observed contexts, not separate rank models.

Pairs

Pair effects are reported as four-lineup contrasts on the log-odds scale: the pair’s coefficient minus the average coefficient of each member with the same-role alternatives to the other, plus the average over alternative pairs. This cancels the hero main effects and each hero’s other relationships and does not depend on where additive hero content is placed. Pairs are shown only with at least 500 identifying matches, that is, distinct development matches in which the signed feature is nonzero. Designated team-ups are labelled; the largest by contrast are Gambit with Rogue (+0.21), Iron Fist with White Fox (+0.20), and Moon Knight with Blade (+0.16). Among pairs that are not designated team-ups, the largest positive contrasts are Gambit with Deadpool (Duelist) (+0.10), Luna Snow with Daredevil (+0.06), and Black Panther with Deadpool (Vanguard) (+0.06) and the largest negative Spider-man with Angela (-0.08), Mantis with Deadpool (Strategist) (-0.07), and Hela with Iron Man (-0.06). The largest opposing matchups are The Thing over Magik (+0.11), Gambit over Angela (+0.08), Phoenix over Elsa Bloodstone (+0.07), and Cloak & Dagger over Venom (+0.07). The standard deviation of the contrasts is 0.050 across team-up-eligible pairs, 0.020 across other allied pairs, and 0.022 across opposing pairs.

A team-up-eligible pair’s coefficient combines ordinary kit complementarity, any mechanic actually used, player selection, coordination, and subsequent decisions; the data do not record activation. A starting-matchup coefficient compares matches that begin with that configuration and includes whatever adaptation follows; it is not the effect of an opponent switching to the counter mid-match.

Stability

The bootstrap stability analysis has not been run for this fit; no stability claim is made.

Confirmation

The locked evaluation compares the candidate and the refitted headline model on the same 19,804 eligible confirmation matches. The paired log-loss improvement is +0.00184 (player-multiplicity bootstrap 95% interval +0.00151 to +0.00217), the candidate’s calibration slope is 1.062 overall, and by rank third 0.908, 1.038, and 1.116. Verdict under the predeclared criterion: PROMOTE.

Table 3: Hero coefficients by lobby rank and same-slot replacement values from the selected development fit

	Coefficient		Replacement, pp		Obs. meta by third						Coefficient		Replacement, pp		Obs. meta by third			
	β_h	slope	Obs. meta	Common	Low	Mid	High	Cov.			β_h	slope	Obs. meta	Common	Low	Mid	High	Cov.
Panel A: Tank (15 heroes)									Panel B: Damage (27 heroes)									
The Thing	+0.177	-0.011	+4.13	+5.02	+4.72	+4.20	+3.56	51	Magik	+0.103	-0.017	+4.07	+4.37	+4.91	+4.09	+3.25	62	
Peni Parker	+0.096	+0.001	+0.95	+2.25	+1.16	+0.97	+0.81	57	Black Cat	+0.128	-0.010	+3.88	+4.17	+4.09	+3.65	+3.39	29	
The Hood	+0.052	+0.021	+1.15	+2.11	+1.05	+1.19	+1.22	41	Storm	+0.131	-0.002	+2.42	+3.03	+2.68	+2.40	+2.17	95	
Magneto	+0.081	+0.026	+0.81	+1.87	+0.55	+0.84	+1.15	41	Daredevil	+0.039	+0.004	+1.95	+2.51	+1.84	+1.92	+2.13	82	
Angela	+0.042	-0.009	+0.14	+1.49	+0.73	+0.09	-0.38	94	Mister Fantastic	+0.088	-0.007	+1.88	+2.51	+2.06	+1.85	+1.74	95	
Hulk	+0.073	-0.007	+0.06	+1.42	+0.75	+0.08	-0.65	90	Iron Fist	+0.055	+0.012	+1.17	+1.76	+0.92	+1.13	+1.47	88	
Thor	+0.049	+0.016	-0.15	+1.21	-0.44	-0.17	+0.14	90	Hela	+0.087	+0.020	+0.70	+1.33	+0.50	+0.75	+0.87	74	
Doctor Strange	-0.025	-0.009	-1.41	+0.06	-0.68	-1.42	-2.10	85	Iron Man	+0.083	-0.020	+0.60	+1.25	+1.52	+0.61	-0.32	93	
Devil Dinosaur	-0.013	+0.006	-1.45	-0.01	-1.21	-1.46	-1.65	90	Winter Soldier	+0.045	-0.015	+0.42	+1.01	+1.01	+0.36	-0.27	56	
Groot	-0.032	+0.006	-1.96	-0.49	-1.84	-1.96	-2.04	83	Moon Knight	+0.040	+0.012	+0.29	+0.95	+0.26	+0.29	+0.31	93	
Venom	-0.040	-0.009	-2.34	-0.83	-1.84	-2.35	-2.77	64	Human Torch	-0.004	-0.020	+0.05	+0.71	+0.83	+0.04	-0.76	93	
Captain America	-0.070	-0.010	-2.28	-0.87	-1.82	-2.30	-2.71	96	Spider-man	-0.003	+0.012	-0.32	+0.41	-0.59	-0.34	-0.07	59	
Emma Frost	-0.098	-0.000	-3.25	-1.55	-2.97	-3.27	-3.49	78	Wolverine	-0.028	+0.036	-0.35	+0.32	-1.24	-0.34	+0.59	94	
Rogue	-0.075	+0.008	-2.97	-1.57	-2.95	-2.99	-2.96	94	Blade	-0.028	-0.021	-0.78	-0.09	-0.04	-0.82	-1.46	94	
Deadpool (Vanguard)	-0.218	-0.031	-6.60	-5.06	-5.45	-6.57	-7.71	95	Black Widow	-0.001	+0.012	-0.91	-0.21	-1.17	-0.92	-0.59	86	
Panel C: Support (13 heroes)									Black Panther	-0.009	-0.036	-1.47	-0.66	+0.03	-1.26	-2.33	48	
Mantis	+0.267	+0.032	+6.44	+6.21	+5.81	+6.59	+7.19	75	Elsa Bloodstone	-0.002	+0.040	-1.24	-0.68	-1.92	-0.78	+0.23	45	
Rocket Raccoon	+0.175	-0.004	+3.87	+3.89	+4.52	+3.92	+3.13	77	Psylocke	-0.018	+0.017	-1.48	-0.77	-1.86	-1.49	-1.06	94	
Ultron	+0.151	-0.023	+3.06	+3.49	+4.11	+3.14	+1.91	95	Hawkeye	-0.052	-0.013	-1.49	-0.79	-0.96	-1.50	-2.03	95	
Adam Warlock	+0.101	-0.033	+1.96	+2.40	+2.95	+1.95	+0.95	95	The Punisher	-0.042	-0.005	-1.77	-1.03	-1.54	-1.78	-1.98	74	
Cloak & Dagger	+0.010	-0.019	+0.29	+0.82	+1.48	+0.47	-0.88	32	Namor	-0.028	-0.015	-1.81	-1.07	-1.03	-1.74	-2.42	68	
Loki	+0.022	+0.035	+0.21	+0.69	-0.40	+0.30	+0.77	94	Deadpool (Duelist)	-0.062	+0.001	-1.86	-1.13	-1.76	-1.87	-1.95	93	
Gambit	+0.030	+0.019	-0.71	-0.31	-1.15	-0.52	-0.10	40	Star-lord	-0.041	-0.006	-2.25	-1.53	-2.15	-2.27	-2.33	94	
Jubilee	-0.066	-0.007	-2.09	-1.26	-1.57	-1.95	-2.65	67	Scarlet Witch	-0.078	-0.002	-2.64	-1.90	-2.25	-2.60	-3.04	94	
Invisible Woman	-0.070	+0.022	-2.12	-1.32	-2.38	-2.08	-1.89	74	Squirrel Girl	-0.146	+0.001	-3.86	-3.12	-3.60	-3.86	-4.12	95	
Jeff The Land Shark	-0.099	+0.018	-2.42	-1.79	-2.52	-2.36	-2.40	92	Phoenix	-0.129	+0.025	-3.98	-3.21	-4.54	-4.00	-3.42	93	
Luna Snow	-0.119	+0.003	-3.70	-2.73	-3.42	-3.65	-4.03	78	Cyclops	-0.128	-0.004	-5.11	-4.12	-4.87	-5.11	-5.33	85	
White Fox	-0.123	-0.016	-3.60	-2.86	-3.01	-3.56	-4.19	89										
Deadpool (Strategist)	-0.279	-0.026	-7.17	-6.37	-6.14	-7.15	-8.23	93										

Notes. β_h is the within-role log-odds contrast at the median lobby score of 4,548, and *slope* its change per standard deviation of lobby rank (252 points), so the coefficient at the 25th and 75th percentiles is $\beta_h \mp$ about $0.67 \times \text{slope}$. Replacement values are exact predicted probability differences in percentage points over 120,000 development matches: observed meta replaces the actual same-role incumbent, common reference compares with the average legal same-role hero in the same slot. Rank thirds are split at lobby scores 4,437 and 4,642. Coverage is the share of same-role slots where the hero was legal (not on that side already, not banned). Heroes are sorted by the common-reference score within role. Cells with fewer than 200 legal contexts are blank.

Table 4: Team-up-eligible starting pairs from the selected fit

Pair	Coef.	Contrast	Matches	Top third %	Pair	Coef.	Contrast	Matches	Top third %
Gambit × Rogue	+0.184	+0.209	22,920	34	Loki × Angela	+0.029	+0.028	3,104	39
Iron Fist × White Fox	+0.188	+0.203	14,182	31	Captain America × Winter Soldier	+0.030	+0.027	7,579	33
Moon Knight × Blade	+0.152	+0.163	4,063	21	The Thing × Iron Fist	+0.021	+0.025	14,162	40
Squirrel Girl × Iron Man	+0.129	+0.139	3,180	13	Human Torch × Invisible Woman	+0.025	+0.024	14,581	38
The Punisher × Rocket Raccoon	+0.125	+0.132	22,765	32	Hulk × Captain America	+0.022	+0.023	2,810	36
Mantis × Star-lord	+0.114	+0.131	16,889	40	Jeff The Land Shark × Devil Dinosaur	+0.026	+0.023	6,651	24
Magik × The Hood	+0.099	+0.109	59,539	34	Wolverine × Cyclops	+0.026	+0.021	3,675	28
Human Torch × Jubilee	+0.099	+0.108	20,113	34	Captain America × Blade	+0.015	+0.021	3,137	26
Ultron × Adam Warlock	+0.140	+0.107	5,927	41	Magneto × Gambit	+0.032	+0.020	62,728	33
Luna Snow × White Fox	+0.100	+0.106	16,956	28	Ultron × Iron Man	+0.018	+0.018	4,055	23
Hulk × Doctor Strange	+0.072	+0.096	8,175	36	Luna Snow × Emma Frost	+0.017	+0.016	34,932	33
Storm × Jeff The Land Shark	+0.087	+0.096	7,545	31	Star-lord × Angela	+0.015	+0.015	2,782	34
Cloak & Dagger × The Hood	+0.076	+0.092	94,581	32	Hulk × Namor	+0.027	+0.015	9,394	30
Doctor Strange × Invisible Woman	+0.091	+0.092	40,971	32	Gambit × Jubilee	-0.006	+0.015	32,597	28
Loki × Mantis	+0.083	+0.090	6,996	48	Cloak & Dagger × Luna Snow	-0.004	+0.014	35,050	33
Jubilee × The Hood	+0.081	+0.086	114,785	30	Phoenix × Rogue	+0.014	+0.013	2,444	26
Rocket Raccoon × Peni Parker	+0.074	+0.083	13,302	32	Psylocke × White Fox	+0.008	+0.013	4,989	25
Mantis × Groot	+0.053	+0.071	13,159	40	Luna Snow × Namor	+0.004	+0.012	14,262	30
Phoenix × Cyclops	+0.069	+0.068	8,596	22	Black Panther × Peni Parker	+0.003	+0.010	2,557	35
Scarlet Witch × Jubilee	+0.060	+0.067	16,017	14	Loki × Hela	+0.001	+0.010	5,953	47
Winter Soldier × Elsa Bloodstone	+0.058	+0.063	16,804	33	Hela × Namor	+0.007	+0.009	10,547	28
Storm × Adam Warlock	+0.040	+0.052	2,124	39	Captain America × Thor	+0.011	+0.007	2,437	24
Mantis × Emma Frost	+0.043	+0.052	28,531	39	Mister Fantastic × The Thing	-0.005	+0.006	5,027	31
Spider-man × Peni Parker	+0.045	+0.050	5,978	33	Cloak & Dagger × Psylocke	+0.006	+0.003	15,160	33
Cloak & Dagger × Moon Knight	+0.049	+0.046	19,222	20	Groot × Jeff The Land Shark	+0.005	+0.002	4,413	29
White Fox × Black Cat	+0.044	+0.046	9,483	18	The Punisher × Winter Soldier	-0.002	+0.000	10,666	34
Venom × Phoenix	+0.045	+0.045	10,303	28	Rocket Raccoon × Squirrel Girl	+0.008	-0.002	6,010	17
Elsa Bloodstone × Devil Dinosaur	+0.037	+0.045	9,421	25	The Punisher × Devil Dinosaur	-0.004	-0.004	6,134	21
Iron Man × Thor	+0.034	+0.044	6,839	24	Human Torch × The Thing	-0.004	-0.004	11,499	36
Magneto × Rogue	+0.028	+0.044	15,160	38	Venom × Spider-man	-0.003	-0.004	19,990	36
Hela × Thor	+0.039	+0.043	10,438	36	Black Panther × Black Cat	+0.000	-0.005	2,662	26
Wolverine × Gambit	+0.038	+0.041	10,108	42	Moon Knight × Elsa Bloodstone	-0.004	-0.006	3,089	18
Hela × Venom	+0.038	+0.041	9,967	41	Mister Fantastic × Invisible Woman	-0.006	-0.007	9,144	30
Hawkeye × Psylocke	+0.037	+0.039	3,584	27	Psylocke × Emma Frost	-0.012	-0.010	8,548	32
Rocket Raccoon × Mister Fantastic	+0.042	+0.039	10,499	29	Venom × Blade	-0.014	-0.010	5,306	26
Venom × Jeff The Land Shark	+0.033	+0.039	8,644	28	Magneto × Emma Frost	-0.027	-0.013	36,898	37
Invisible Woman × The Thing	+0.050	+0.039	29,386	38	Rocket Raccoon × Groot	-0.012	-0.016	13,270	32
Hela × Phoenix	+0.039	+0.037	4,617	36	Thor × Angela	-0.011	-0.017	2,685	32
Doctor Strange × Magik	+0.042	+0.037	24,603	33	Hulk × Wolverine	-0.020	-0.017	4,536	43
Mantis × Adam Warlock	+0.069	+0.035	4,275	46	Magneto × Scarlet Witch	-0.008	-0.018	18,147	18
Squirrel Girl × Spider-man	+0.030	+0.035	3,125	20	Blade × Jubilee	-0.016	-0.019	14,149	22
The Punisher × Daredevil	+0.030	+0.034	3,862	39	Luna Snow × Adam Warlock	+0.009	-0.029	5,084	44
Hawkeye × Cloak & Dagger	+0.041	+0.033	13,883	25	Hawkeye × Black Widow	-0.028	-0.029	2,174	31
Groot × Star-lord	+0.033	+0.033	5,188	33	Storm × Human Torch	-0.032	-0.030	2,461	34
Doctor Strange × Scarlet Witch	+0.038	+0.033	8,960	16	Gambit × Cyclops	-0.034	-0.030	20,331	27
Black Widow × Phoenix	+0.032	+0.032	4,041	35	Storm × Thor	-0.029	-0.031	5,756	27
Storm × Black Panther	+0.033	+0.031	2,787	32	Hulk × Iron Man	-0.033	-0.032	4,611	27
Black Widow × Daredevil	+0.027	+0.030	5,429	50	Iron Fist × Daredevil	-0.049	-0.046	5,253	42
Ultron × Peni Parker	+0.015	+0.029	2,422	29	Black Panther × Magik	-0.047	-0.049	6,568	39
Star-lord × Adam Warlock	+0.015	+0.029	2,006	35	Spider-man × Black Cat	-0.054	-0.057	7,537	24

Notes. Allied pairs designated as team-ups in the Season 9.5 catalogue, with at least 500 identifying matches, sorted by contrast. *Coef.* is $S_{hk}(r)$ at the median lobby rank under the uniform centering convention (each hero's allied deviations sum to zero across its partners and within each role pair). *Contrast* is the four-lineup log-odds contrast against same-role alternatives. *Matches* counts distinct development matches with a nonzero signed feature; *Top third %* is the share of those matches in the top lobby-rank third. Read as the synergy of a team-up-eligible starting pair.

Table 5: Other allied pairs: largest positive and negative contrasts

<i>Largest positive</i> Pair	Coef.	Contrast	Matches	Top third %	<i>Largest negative</i> Pair	Coef.	Contrast	Matches	Top third %
Gambit × Deadpool (Duelist)	+0.090	+0.103	13,919	37	Spider-man × Angela	-0.067	-0.075	6,072	38
Luna Snow × Daredevil	+0.055	+0.059	14,834	46	Mantis × Deadpool (Strategist)	-0.046	-0.071	3,888	36
Black Panther × Deadpool (Vanguard)	+0.047	+0.058	2,435	30	Hela × Iron Man	-0.052	-0.060	2,659	29
Mister Fantastic × Rogue	+0.044	+0.057	1,897	28	Doctor Strange × Magneto	-0.076	-0.059	22,879	37
Groot × Emma Frost	+0.043	+0.056	4,985	31	Doctor Strange × Mister Fantastic	-0.053	-0.057	4,850	27
Angela × The Hood	+0.052	+0.053	10,573	35	Ultron × Devil Dinosaur	-0.056	-0.056	2,440	22
Jeff The Land Shark × Deadpool (Strategist)	+0.043	+0.052	2,990	23	Star-lord × Invisible Woman	-0.052	-0.056	9,848	34
Jeff The Land Shark × Invisible Woman	+0.026	+0.050	9,018	30	Human Torch × White Fox	-0.045	-0.055	3,889	26
The Punisher × Invisible Woman	+0.044	+0.049	15,011	33	Emma Frost × Gambit	-0.035	-0.053	26,315	32
Hulk × Deadpool (Strategist)	+0.038	+0.049	5,149	33	Jubilee × Rogue	-0.038	-0.051	11,643	25
Storm × Devil Dinosaur	+0.039	+0.046	2,855	25	Adam Warlock × Deadpool (Strategist)	-0.006	-0.051	1,187	30
Loki × Psylocke	+0.035	+0.046	2,088	44	Loki × Adam Warlock	-0.011	-0.050	818	46
Loki × Gambit	+0.028	+0.046	6,326	45	Hela × Ultron	-0.053	-0.049	3,179	37
Luna Snow × Psylocke	+0.037	+0.046	8,908	40	Magik × The Thing	-0.043	-0.049	14,601	38
Magneto × Wolverine	+0.039	+0.046	16,482	48	Namor × Gambit	-0.049	-0.049	13,436	29
Loki × Cloak & Dagger	+0.021	+0.046	8,641	37	Wolverine × The Thing	-0.052	-0.048	4,489	45
Magik × Star-lord	+0.035	+0.045	4,706	35	Peni Parker × Gambit	-0.035	-0.047	6,877	32
Venom × Elsa Bloodstone	+0.040	+0.045	6,936	32	Mantis × Devil Dinosaur	-0.046	-0.047	7,705	34
Jeff The Land Shark × Deadpool (Duelist)	+0.036	+0.044	4,462	28	Human Torch × Jeff The Land Shark	-0.039	-0.047	3,374	32
Mantis × Elsa Bloodstone	+0.036	+0.043	10,927	39	Daredevil × Rogue	-0.043	-0.045	3,897	40
Angela × Devil Dinosaur	+0.037	+0.042	1,461	28	Loki × Ultron	-0.018	-0.045	651	33
Captain America × Hela	+0.040	+0.042	2,843	37	Iron Fist × Jubilee	-0.034	-0.045	21,512	33
Namor × Invisible Woman	+0.031	+0.042	15,997	28	The Punisher × Loki	-0.041	-0.044	3,226	38
Devil Dinosaur × Deadpool (Strategist)	+0.037	+0.042	4,645	20	Adam Warlock × White Fox	+0.005	-0.043	1,735	23
Magik × Deadpool (Strategist)	+0.043	+0.041	9,146	29	Squirrel Girl × Cyclops	-0.035	-0.043	2,385	11

Notes. Allied pairs that are not designated team-ups, with at least 500 identifying matches. Columns as in Table 4. A negative contrast means the pair does worse together than its members' other pairings predict.

Table 6: Opposing matchups: largest contrasts

Matchup	Coef.	Contrast	Matches	Top third %	Matchup	Coef.	Contrast	Matches	Top third %
The Thing over Magik	+0.085	+0.106	16,054	38	Black Panther over Mantis	+0.053	+0.053	9,327	46
Gambit over Angela	+0.066	+0.075	8,342	36	Mister Fantastic over Black Widow	+0.048	+0.053	2,976	35
Phoenix over Elsa Bloodstone	+0.065	+0.071	3,643	24	Rocket Raccoon over Squirrel Girl	+0.045	+0.052	6,015	17
Cloak & Dagger over Venom	+0.057	+0.069	28,213	31	Namor over Gambit	+0.044	+0.052	13,764	29
Wolverine over Cloak & Dagger	+0.051	+0.067	12,632	38	Gambit over Daredevil	+0.047	+0.051	13,091	41
Spider-man over Iron Man	+0.064	+0.066	5,705	26	Mister Fantastic over Black Cat	+0.047	+0.051	2,967	21
Black Widow over Iron Man	+0.063	+0.065	3,873	29	Black Widow over Hela	+0.045	+0.051	9,464	48
The Thing over Deadpool (Duelist)	+0.056	+0.063	6,190	36	Captain America over Invisible Woman	+0.049	+0.050	7,080	31
Magik over The Hood	+0.063	+0.063	53,481	34	Peni Parker over Magik	+0.033	+0.050	5,887	32
Namor over Spider-man	+0.061	+0.063	8,682	32	The Thing over Rogue	+0.054	+0.050	5,708	34
Magik over Devil Dinosaur	+0.065	+0.062	10,315	27	Iron Fist over Black Widow	+0.042	+0.050	7,097	46
Black Widow over Ultron	+0.060	+0.061	2,983	38	Mister Fantastic over Magneto	+0.038	+0.050	14,966	34
The Punisher over The Thing	+0.057	+0.061	8,725	34	Winter Soldier over Hulk	+0.038	+0.048	17,460	43
Ultron over Magik	+0.049	+0.061	5,854	32	Spider-man over Rocket Raccoon	+0.043	+0.048	26,191	37
Deadpool (Duelist) over Invisible Woman	+0.055	+0.060	12,725	35	Spider-man over Human Torch	+0.046	+0.048	7,436	40
Iron Fist over Rocket Raccoon	+0.053	+0.060	16,582	40	Hulk over Wolverine	+0.040	+0.048	3,531	47
Storm over Iron Fist	+0.060	+0.060	3,805	38	The Thing over Mister Fantastic	+0.038	+0.048	4,781	31
Black Cat over Mantis	+0.052	+0.058	7,919	28	Psylocke over Moon Knight	+0.045	+0.048	2,397	23
Venom over Squirrel Girl	+0.050	+0.056	3,247	15	Iron Fist over Invisible Woman	+0.037	+0.048	18,123	40
Daredevil over Black Cat	+0.049	+0.056	4,371	30	Daredevil over Rocket Raccoon	+0.043	+0.047	14,532	45
Hawkeye over Storm	+0.048	+0.055	1,684	29	The Hood over The Punisher	+0.045	+0.047	28,208	31
Daredevil over Invisible Woman	+0.044	+0.054	16,320	44	Emma Frost over Black Widow	+0.046	+0.047	13,188	38
Invisible Woman over Winter Soldier	+0.049	+0.054	47,155	40	Devil Dinosaur over Gambit	+0.041	+0.047	12,794	27
Captain America over Spider-man	+0.050	+0.054	4,608	32	Jubilee over Iron Fist	+0.039	+0.047	23,217	32
Magneto over Hela	+0.059	+0.053	40,218	46	Gambit over Jeff The Land Shark	+0.033	+0.047	12,936	27

Notes. A over B means A 's side wins more often when the two start on opposite sides than the two heroes' other matchups predict; the coefficient and contrast are oriented so that both are positive for A . At least 500 identifying matches. Starting matchups include whatever adaptation followed.